

Suggested Revisions: Phase-Normalized Streaming EEG Artifact Rejection

Summary

The manuscript has a strong practical idea, but the revision should shift the emphasis away from simple heartbeat-triggered averaging and toward a **phase-normalized, streaming disturbance-rejection framework for wearable EEG**. The current offline method estimates a cardiac artifact template by averaging EEG samples at fixed offsets from detected R-peaks. This is useful, but closely related to conventional heartbeat-locked template subtraction. The revision should therefore emphasize the phase normalization, streaming implementation, and timing-only disturbance-rejection formulation.

1 Move from Offline Fixed-Time Averaging to Phase-Normalized Streaming Cancellation

The strongest revision would be to make **normalized cardiac phase** the central representation of the algorithm.

For heartbeat k , define

$$\phi_k(t) = \frac{t - t_k}{t_{k+1} - t_k}, \quad 0 \leq \phi_k < 1. \quad (1)$$

Each heartbeat is then resampled onto a common normalized phase grid. The cardiac artifact template is learned in phase space rather than at fixed sample offsets:

$$\hat{d}_k(\phi) = \hat{d}_{k-1}(\phi) + a \left[y_k(\phi) - \hat{d}_{k-1}(\phi) \right]. \quad (2)$$

For each new heartbeat, the learned phase template is warped back to the current RR interval and subtracted from the EEG:

$$\hat{s}(t) = y(t) - \hat{d}_k \left(\frac{t - t_k}{t_{k+1} - t_k} \right). \quad (3)$$

This makes the repetitive structure periodic in *cardiac phase*, even when it is not periodic in clock time due to heart-rate variability.

The proposed streaming pipeline becomes

R -peak detection $\rightarrow$ RR estimation $\rightarrow$ phase normalization $\rightarrow$ template update $\rightarrow$ inverse phase warping $\rightarrow$ EEG

This formulation also provides a clearer link to repetitive control and iterative learning control.

2 Synthetic Example Demonstrating the Benefit of Phase Normalization

A synthetic example was generated to explicitly test heart-rate variability. Clean EEG was contaminated with a cardiac artifact whose morphology was constant in normalized cardiac phase but whose duration stretched or compressed according to the RR interval.

The simulated RR intervals had approximately

$$\text{mean} = 0.854 \text{ s}, \quad SD = 0.095 \text{ s},$$

with a range of approximately 0.625–1.15 s.

Two methods were compared:

1. conventional fixed R-peak-locked averaging;
2. normalized cardiac-phase averaging.

The results were:

Method	CFA removal	Clean EEG RMSE	Clean EEG corr.	CFA corr.
Fixed R-locked template	10.63 dB	1.30 μV	0.968	0.956
Normalized-phase template	18.31 dB	0.54 μV	0.994	0.993

Table 1: Synthetic comparison of fixed-time and normalized-phase cardiac artifact removal.

Thus, phase normalization provided approximately

$$18.31 - 10.63 = 7.68 \text{ dB}$$

additional artifact suppression in this example.

The mechanism is intuitive. When beats are aligned only by the R-peak, later components such as the T-wave-related field occur at different absolute times as RR duration changes. Averaging therefore smears the artifact waveform. After converting each beat to normalized phase, the artifact components align more closely.

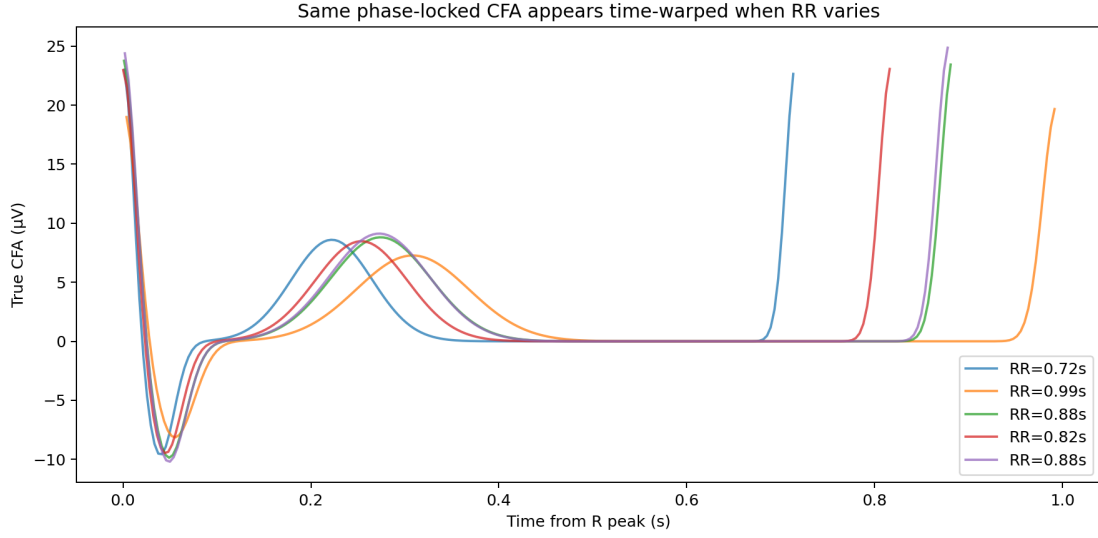


Figure 1: Synthetic cardiac artifact under heart-rate variability. The same phase-locked morphology appears stretched or compressed in ordinary time as RR duration changes.

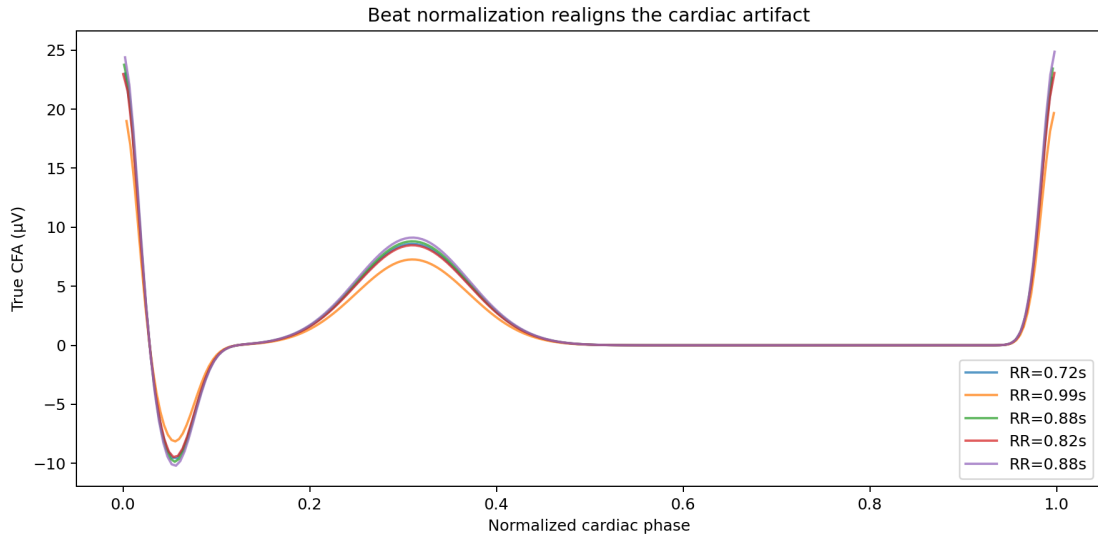


Figure 2: The same synthetic beats after normalization to cardiac phase. Phase normalization realigns the artifact morphology across variable RR intervals.

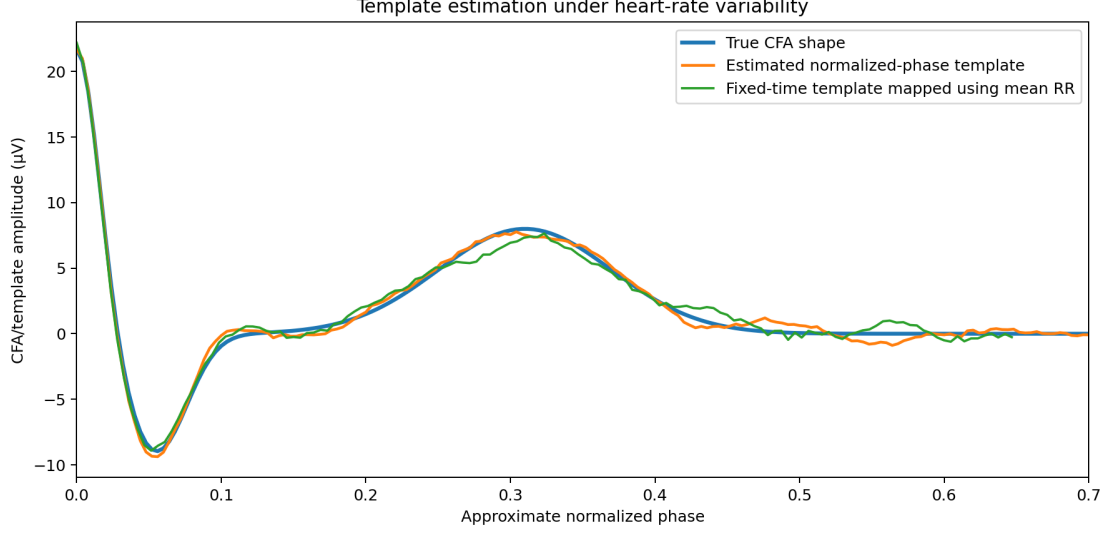


Figure 3: Comparison of the true cardiac artifact morphology with the estimated normalized-phase template and the fixed-time template mapped using the mean RR interval.

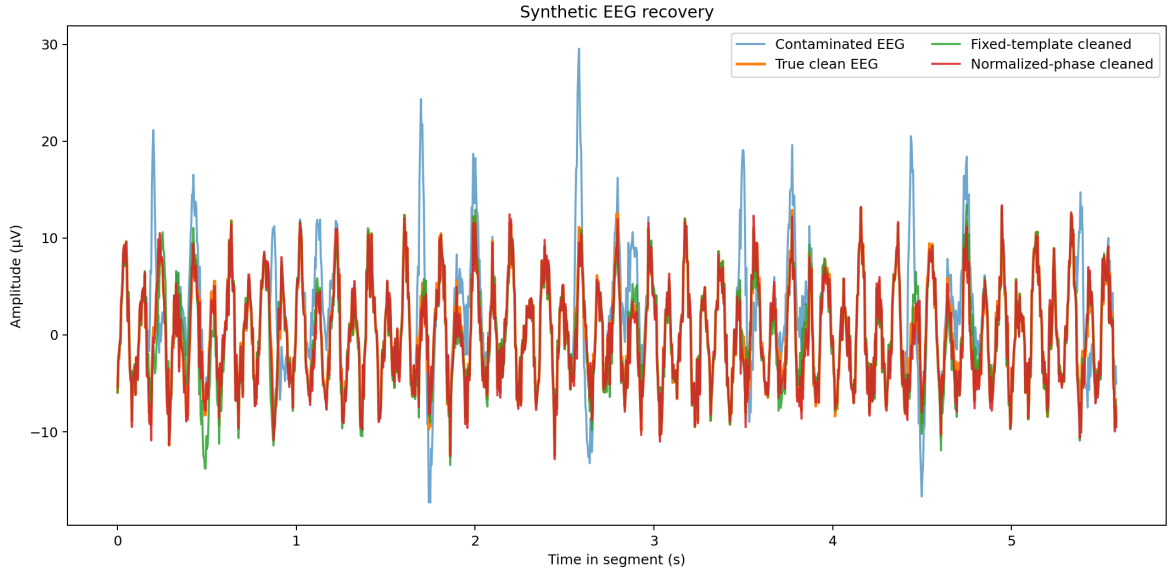


Figure 4: Representative synthetic EEG segment showing the contaminated signal, true clean EEG, fixed-template reconstruction, and normalized-phase reconstruction.

3 Reframe the Contribution Around Cardiac Field Artifact Removal

The paper should avoid presenting heartbeat-triggered averaging itself as the primary algorithmic novelty. A stronger framing is:

We formulate cardiac-field artifact suppression as a heartbeat-phase disturbance-rejection

problem. Unlike conventional ECG-reference regression, the framework uses the cardiac sensor mainly as a timing source and learns the artifact directly from EEG in cardiac-phase coordinates.

The key conceptual point is

heartbeat timing can remain informative even when ECG-to-EEG waveform coherence is weak

The target operating regime is therefore

low-channel wearable EEG + reliable heartbeat timing + weak waveform predictability

This framing is more defensible than claiming novelty from the averaging operation itself.

4 Strengthen the Streaming Contribution

The revised manuscript should directly compare

offline non-causal vs. one-beat-delayed streaming vs. predicted-phase streaming.

A useful simulation would vary R-peak prediction error:

0, 2, 5, 10, 20, 40 ms,

and quantify CFA suppression as prediction jitter increases.

This would make the paper more clearly a real-time control contribution rather than only a preprocessing method.

5 Add a Systematic Heart-Rate-Variability Experiment

The normalized-phase benefit should be tested systematically as a function of RR variability:

$\sigma_{RR} = 0, 20, 40, 60, 80, 100, 150$ ms.

For each level, compare fixed-time and normalized-phase cancellation.

The expected pattern is that both methods perform similarly when RR variability is near zero, while normalized-phase cancellation increasingly outperforms fixed-time averaging as heart-rate variability increases.

This would provide a direct mechanistic validation of the proposed algorithm.

6 Include Appropriate Baseline Methods

The revised manuscript should compare the proposed method against:

- fixed heartbeat-triggered average template;
- sliding-window heartbeat template;
- exponentially updated heartbeat template;
- ECG-reference RLS adaptive cancellation;
- normalized-phase streaming method.

This is important because the proposed ILC update can mathematically resemble exponential template averaging. The novelty should therefore be demonstrated at the level of the complete phase-normalized streaming architecture.

7 Clarify Preservation of Heartbeat-Related Neural Activity

A genuine heartbeat-evoked potential can be phase locked to the cardiac cycle. Therefore, a phase-locked template subtraction method may remove some neural activity together with the artifact.

The method should be positioned primarily for applications where heartbeat-correlated activity is treated as nuisance contamination, especially continuous wearable EEG. It should not automatically be used in studies where heartbeat-evoked cortical responses are themselves the biological signal of interest.

8 Future Extension: Streaming Eye-Blink Removal

A natural extension is eye-blink artifact cancellation.

Eye blinks are not periodic like heartbeats, but they are stereotyped transient events whose duration varies substantially. If blink onset, peak, and offset can be detected from EOG or frontal EEG, each blink can be normalized into an event-phase representation.

A useful mapping is

$$\text{blink onset} \rightarrow 0, \quad \text{blink peak} \rightarrow 0.5, \quad \text{blink offset} \rightarrow 1.$$

For EEG channel j , define a blink artifact template:

$$\hat{d}_{\text{blink}}^{(j)}(\phi) = \frac{1}{N} \sum_{k=1}^N EEG_k^{(j)}(\phi). \quad (4)$$

This template can then be warped back to the actual duration of each detected blink and subtracted in streaming operation.

The resulting pipeline is

$$EOG \rightarrow \text{blink detection} \rightarrow \{\text{onset, peak, offset}\} \rightarrow \text{phase normalization} \rightarrow \text{adaptive template} \rightarrow \text{EEG cleaning}$$

This extends the cardiac formulation into a broader class of **event-phase adaptive artifact rejection** methods.

Overall Recommended Framing

The manuscript can be repositioned around the broader idea

$$\text{physiological event timing} \rightarrow \text{normalized phase coordinate} \rightarrow \text{adaptive disturbance learning} \rightarrow \text{streaming EEG a}$$

Heartbeat removal is the first application, where ECG or PPG supplies the timing reference. Eye-blink removal can be presented as a future extension, where EOG supplies the event reference.

The most important revision is therefore not simply to improve the offline cardiac template, but to make **phase-normalized streaming disturbance rejection** the central methodological contribution.